ICS Assignment Submission

Q1:

Input	Processing	Output
packageWeight repeat pickupPC deliveryPC	• Prompt for packageWeight • Get packageWeight • Prompt for pickupPC • Get pickupPC • Prompt for deliveryPC • Get deliveryPc • Calculate deliveryCharge • Display deliveryCharge • Display • Prompt for repeat • Get repeat	deliveryCharge message

[Pseudocode]:

```
Display_deliveryCharge
1 SET message = "Do you want to calculate for another delivery(Yes/No)?"
2 SET repeat = “”
3 DOWHILE (repeat != “no”)
4 Prompt for packageWeight
5 Get packageWeight
6 IF (packageWeight <= 5 AND packageWeight > 0)
      Prompt for pickupPC
      Get pickupPC

  IF (pickupPC >= 10000 && pickupPC < 90000) OR
     (pickupPC >= 170000 && pickupPC < 200000) OR
     (pickupPC >= 220000 && pickupPC < 240000)
      pickupInCBD = true
  ELSEIF (pickupPC < 9999)
      Display = “The postal code you provided was invalid. Please provide a valid postal code for pickup: ”
  ELSE
      pickupInCBD = false
  ENDIF

  IF (deliveryPC >= 10000 && deliveryPC< 90000) OR
     (deliveryPC >= 170000 && deliveryPC< 200000) OR
     (deliveryPC >= 220000 && deliveryPC< 240000)
```

```
        deliveryInCBD = true


    ELSEIF (deliveryPC < 9999)
        DISPLAY = "The postal code you provided was invalid. Please provide a valid postal code for delivery:
"
    ELSE
        deliveryInCBD = false
    ENDIF
ELSE
    DISPLAY = "Invalid Weight"
ENDIF


7 IF (pickupInCBD == true AND deliveryInCBD == true)
        deliveryCharge = 15
        DISPLAY deliveryCharge
    ELSEIF (pickupInCBD == false AND deliveryInCBD == false)
        deliveryCharge = 25
        DISPLAY deliveryCharge
    ELSE
        deliveryCharge = 20
        DISPLAY deliveryCharge
    ENDIF

8 Prompt for repeat
9 GET repeat
 ENDDO
    END
```

[Desk Checking]

1. Input Data

	First Data Set	Second Data Set	Third Data Set
packageWeight	3	5	7
pickupPC	16400	110000	123456
deliveryPC	180111	180111	153000
repeat	yes	no	yes

2. Expected Results

	First Data Set	Second Data Set	Third Data Set
deliveryCharge	15	20	
message	repeat	repeat	repeat
display			Invalid Weight

3. Desk Check Table

Statement No	package Weight	pickupPC	deliveryPC	deliveryCharge	message	repeat	display	DOWHILE
First Pass								
1					Do you want to calculate for another delivery(Yes/No)?			
2						Blank		
3								TRUE

4,5	3							
6		16400	180111					
7				15				
8,9						yes		
3								TRUE
4,5	5							
6		110000	180111					
7				20				
8,9						yes		
3								TRUE
4,5	7							
6							Invalid Weigh t	
7								
8,9						no		

Test Case 1:

```
Output - Delivery Algorithm (run)
run:
Please provide the weight of your parcel [0-5kg]: 3
Please provide a 6-digit postal code for pickup: 016400
Please provide a 6-digit postal code for delivery: 180111
Delivery Cost: $15
Do you want to calculate for another delivery(Yes/No)?
yes
Please provide the weight of your parcel [0-5kg]:
```

Test Case 2:

```
Output - Delivery Algorithm (run)
run:
Please provide the weight of your parcel [0-5kg]: 5
Please provide a 6-digit postal code for pickup: 110000
Please provide a 6-digit postal code for delivery: 180111
Delivery Cost: $20
Do you want to calculate for another delivery(Yes/No)?
yes
Please provide the weight of your parcel [0-5kg]:
Terminal - localhost   Output
```

Test Case 3:

```
Output - Delivery Algorithm (run)
run:
Please provide the weight of your parcel [0-5kg]: 7
Invalid Weight. Cannot proceed with delivery.
Do you want to calculate for another delivery(Yes/No)?
```

```java
import java.util.Scanner;

public class DeliveryAlgorithm {

    public static void main(String[] args) {
        Scanner scanner = new Scanner(source: System.in);
        String message = "Do you want to calculate for another delivery(Yes/No)?";
        String repeat = "";
        int deliveryCharge;

        while (!repeat.equalsIgnoreCase(anotherString: "no")){
            System.out.print(s: "Please provide the weight of your parcel [0-5kg]: ");
            double packageWeight = scanner.nextDouble();

            if (packageWeight <= 5.0 && packageWeight > 0){
                System.out.print(s: "Please provide a 6-digit postal code for pickup: ");
                // pickupPC = Pickup Postal Code
                int pickupPC = scanner.nextInt();
                boolean pickupInCBD;

                while (pickupPC <= 9999){
                    System.out.println(x: "The postal code you provided was invalid.");
                    System.out.print(s: "Please provide a valid postal code for pickup: ");
                    pickupPC = scanner.nextInt();
                }

                if(pickupPC >= 10000 && pickupPC < 90000 || pickupPC >= 170000 && pickupPC < 200000
                        || pickupPC >= 220000 && pickupPC < 240000)
                    pickupInCBD = true;
                else
                    pickupInCBD = false;

                System.out.print(s: "Please provide a 6-digit postal code for delivery: ");
                //deliveryPC = Pickup Postal Code
                int deliveryPC = scanner.nextInt();
                boolean deliveryInCBD;

                while (deliveryPC <= 9999){
                    System.out.print(s: "Invalid, please provide valid postal code: ");
                    deliveryPC = scanner.nextInt();
                }

                if(deliveryPC >= 10000 && deliveryPC < 90000 ||
                        deliveryPC >= 170000 && deliveryPC < 200000 ||
                        deliveryPC >= 220000 && deliveryPC < 240000)
                    deliveryInCBD = true;
                else
                    deliveryInCBD = false;

                if (pickupInCBD == true && deliveryInCBD == true){
                    deliveryCharge = 15;
                    System.out.println("Delivery Cost: $" + deliveryCharge);
                }
                else if (pickupInCBD == false && deliveryInCBD == false){
                    deliveryCharge = 25;
                    System.out.println("Delivery Cost: $" +deliveryCharge);
                }
                else {
                    deliveryCharge = 20;
                    System.out.println("Delivery Cost: $" +deliveryCharge);
                }
            }

            else {
                System.out.println(x: "Invalid Weight. Cannot proceed with delivery.");
            }

          while (true){
              System.out.println(x: message);
              repeat = scanner.next();
              if (repeat.equalsIgnoreCase(anotherString: "yes") || repeat.equalsIgnoreCase(anotherString: "n
                  break;}
              else {
                  System.out.println(x: "Invalid answer. Please enter 'yes' or 'no'.");
              }
          }
```

Algorithm:

```
import java.util.Scanner;

public class DeliveryAlgorithm {

    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        String message = "Do you want to calculate for another delivery(Yes/No)?";
        String repeat = "";
        int deliveryCharge;

        while (!repeat.equalsIgnoreCase("no")){
            System.out.print("Please provide the weight of your parcel [0-5kg]: ");
            double packageWeight = scanner.nextDouble();

            if (packageWeight <= 5.0 && packageWeight > 0){
                System.out.print("Please provide a 6-digit postal code for pickup: ");
                // pickupPC = Pickup Postal Code
                int pickupPC = scanner.nextInt();
                boolean pickupInCBD;

                while (pickupPC <= 9999){
                    System.out.println("The postal code you provided was invalid.");
                    System.out.print("Please provide a valid postal code for pickup: ");
                    pickupPC = scanner.nextInt();
                }

                if(pickupPC >= 10000 && pickupPC < 90000 || pickupPC >= 170000 && pickupPC < 200000
                        || pickupPC >= 220000 && pickupPC < 240000)
```

```
    pickupInCBD = true;
else
    pickupInCBD = false;

System.out.print("Please provide a 6-digit postal code for delivery: ");
//deliveryPC = Pickup Postal Code
int deliveryPC = scanner.nextInt();
boolean deliveryInCBD;

while (deliveryPC <= 9999){
    System.out.print("Invalid, please provide valid postal code: ");
    deliveryPC = scanner.nextInt();
}

if(deliveryPC >= 10000 && deliveryPC < 90000 ||
        deliveryPC >= 170000 && deliveryPC < 200000 ||
        deliveryPC >= 220000 && deliveryPC < 240000)
    deliveryInCBD = true;
else
    deliveryInCBD = false;

if (pickupInCBD == true && deliveryInCBD == true){
    deliveryCharge = 15;
    System.out.println("Delivery Cost: $" + deliveryCharge);
}
else if (pickupInCBD == false && deliveryInCBD == false){
    deliveryCharge = 25;
    System.out.println("Delivery Cost: $" +deliveryCharge);
}
```

```
            else {
                deliveryCharge = 20;
                System.out.println("Delivery Cost: $" +deliveryCharge);
            }
        }

        else {
            System.out.println("Invalid Weight. Cannot proceed with delivery.");
        }

        while (true){
            System.out.println(message);
            repeat = scanner.next();
            if (repeat.equalsIgnoreCase("yes") || repeat.equalsIgnoreCase("no")) {
                break;}
            else {
                System.out.println("Invalid answer. Please enter 'yes' or 'no'.");
            }
        }
    }
  }
}
```

Code Logic:

This code is a simple delivery cost calculator. It prompts the user for the weight of a parcel and postal codes for pickup and delivery. It then determines whether both locations are within the Central Business District (CBD) or not. Based on these conditions, it calculates the delivery charge: $15 for CBD-to-CBD, $25 for non-CBD-to-non-CBD, and $20 for other combinations. If the parcel's weight is outside the valid range (0-5kg), it displays an error message. The code repeats the process until the user chooses to stop. It demonstrates how conditional statements can be used to make decisions based on user input and

geographic criteria, allowing for the calculation of appropriate delivery charges based on the pickup and delivery locations.

Q2:

Input	Processing	Output
Read Taxpayer's Record • ID • Name • Income	• Read record from file • Calculate incomeTax • Display Name, incomeTax	• Name • Income Tax • Msg

[Pseudocode]:

```
Display_IncomeTax
    1.  SET IncomeTax
    2.  SET msg = ""
    3.  Read first record
    4.  DOWHILE NOT EOF (End of File)
    5.  IF (income < 0)
          Msg = "Invalid Income"

        ELSE IF (income < 0 && income >=20000)
          incomeTax = 0

        ELSE IF (income < 20000 && income >=40000)
          taxableIncome = income - 20000
          incomeTax = (taxableIncome*0.02)

        ELSE IF (income < 40000 && income >=70000)
          taxableIncome = income - 40000
          incomeTax = 400 + (taxableIncome*0.035)

          ELSE IF (income < 70000 && income >=110000)
          taxableIncome = income - 70000
          incomeTax = 400 + 1050 +(taxableIncome*0.07)

           ELSE
```

```
taxableIncome = income - 110000
incomeTax = 400 + 1050 + 2800 + taxableIncome * 0.115
 ENDIF
ENDIF
ENDIF
ENDIF
ENDIF
ENDIF
```

6. Display Name, incomeTax,msg

7. READ next record

ENDDO

[Desk Checking]:

1. Input Data

	First Record	Second Record	Third Record	Fourth Record
ID	A0001	B0002	C0003	D0004
Name	Chandler Bing	Ross Geller	Joey Tribbiani	Gunther
Income	65,000.00	95,000.00	-132,000.00	150,000.00

2. Expected Results

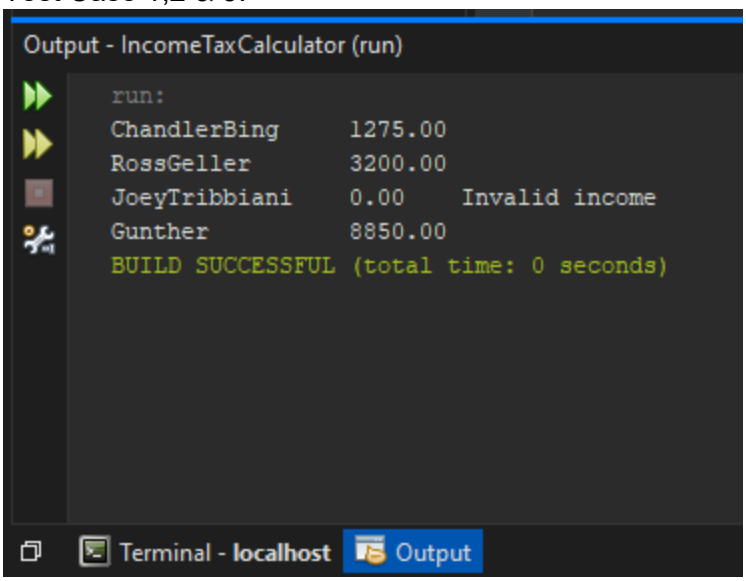
	First Data Set	Second Data Set	Third Data Set	Fourth Data Set
Name	Chandler Bing	Ross Geller	Joey Tribbiani	Gunther
Income Tax	1,275	3,200	0	26,100
Msg			Invalid Income	

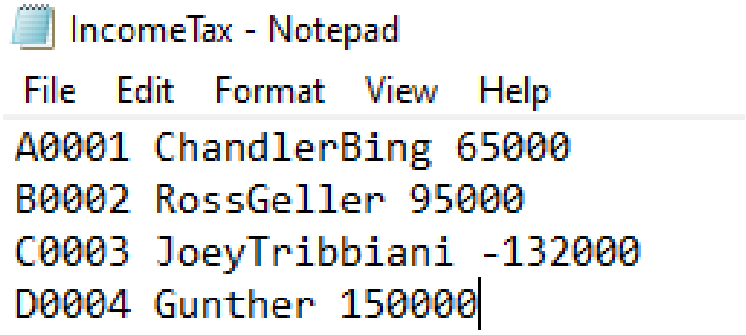
3. Desk Check Table

Statement No.	ID	Name	Income	Tax Amount	DOWHILE	Msg
First Pass						
1,2,3,4						
5	A0001	Chandler Bing	65000			
6		Chandler Bing		1275		
7					NOT EOF	
5	B0002	Ross Geller	95000			
6		Ross Geller		3200		
7					NOT EOF	
5	C0003	Joey Tribbiani	-132000			
6		Joey Tribbiani		0		Invalid Income
7					NOT EOF	

5	D0004	Gunther	150000			
6		Gunther		8850		
7					End	

Test Case 1,2 & 3:





Output - IncomeTaxCalculator (run)

```
run:
SheldonCooper            2150.00
PennyHofstadter          3550.00
LeonardHofstadter        1800.00
HowardWolowitz           1450.00
AmyCooper                2500.00
RajKoothrappali          1800.00
BernadetteWolowitz       2850.00
StuartBloom              750.00
EmilySweeney             1100.00
BarryKripke              1450.00
BUILD SUCCESSFUL (total time: 0 seconds)
```

IncomeTax - Notepad

File Edit Format View Help

```
A0001 SheldonCooper       80000
B0002 PennyHofstadter     100000
C0003 LeonardHofstadter   75000
D0004 HowardWolowitz      70000
E0005 AmyCooper           85000
F0006 RajKoothrappali     75000
G0007 BernadetteWolowitz  90000
H0008 StuartBloom         50000
I0009 EmilySweeney        60000
J0010 BarryKripke         70000
```

Output - IncomeTaxCalculator (run)

```
run:
HarveySpecter    5400.00
MikeRoss         3200.00
LouisLitt        7125.00
BUILD SUCCESSFUL (total time: 0 seconds)
```

IncomeTax - Notepad

File Edit Format View Help

```
A0001 HarveySpecter 120000
B0002 MikeRoss      950000
C0003 LouisLitt     135000
```

```java
import java.util.Scanner;
import java.io.*;
public class IncomeTaxCalculator {

    public static void main(String[] args) throws Exception {
        Scanner input = new Scanner(new File("D:\\Study Materials\\Kaplan\\Term 3\\Introduction to Computer Science\\IncomeTaxCalculator\\IncomeTax.txt"));
        String employeeID,employeeName, msg;
        double income,taxableIncome,incomeTax;
        while(input.hasNext()){
            incomeTax = 0;
            msg = "";
            employeeID = input.next();
            employeeName = input.next();
            income = input.nextDouble();
            if(income < 0) {
                msg ="Invalid income";
            }

            else if(income >0 && income <=20000){
                incomeTax = 0;
            }

            else if(income >20000 && income <=40000){
                taxableIncome = income - 20000;
                incomeTax = (taxableIncome*0.02);
            }

            else if(income >40000 && income <=70000){
                taxableIncome = income - 40000;
                incomeTax = 400 + (taxableIncome*0.035);
            }

            else if(income >70000 && income <=110000){
                taxableIncome = income - 70000;
                incomeTax = 400 + 1050 +(taxableIncome*0.07);
            }

            else{
                taxableIncome = income - 110000;
                incomeTax = 400 + 1050 + 2800 + taxableIncome * 0.115;
            }

            System.out.printf("%s \t %.2f  \t %s\n",
                    employeeName, incomeTax, msg);
        }
    }
}
```

Algorithm:

```java
import java.util.Scanner;
import java.io.*;
public class IncomeTaxCalculator {

   public static void main(String[] args) throws Exception {
      Scanner input = new Scanner(new File("D:\\Study Materials\\Kaplan\\Term 3\\Introduction to
Computer Science\\IncomeTaxCalculator\\IncomeTax.txt"));
      String employeeID,employeeName, msg;
      double income,taxableIncome,incomeTax;
      while(input.hasNext()){
         incomeTax = 0;
         msg = "";
         employeeID = input.next();
         employeeName = input.next();
         income = input.nextDouble();
         if(income < 0) {
            msg ="Invalid income";
         }

         else if(income >0 && income <=20000){
```

```
            incomeTax = 0;
        }

        else if(income >20000 && income <=40000){
            taxableIncome = income - 20000;
            incomeTax = (taxableIncome*0.02);
        }

        else if(income >40000 && income <=70000){
            taxableIncome = income - 40000;
            incomeTax = 400 + (taxableIncome*0.035);
        }

        else if(income >70000 && income <=110000){
            taxableIncome = income - 70000;
            incomeTax = 400 + 1050 +(taxableIncome*0.07);
        }

        else{
            taxableIncome = income - 110000;
            incomeTax = 400 + 1050 + 2800 + taxableIncome * 0.115;
        }

        System.out.printf("%s \t %.2f  \t %s\n",
                employeeName, incomeTax, msg);
    }
  }
}
```

Code Logic:

This code calculates income tax for employees based on their income. It reads employee data from a file, including employee ID, name, and income. The code then uses a series of conditional statements to determine the income tax based on income brackets. For income values less than or equal to $20,000, the income tax is $0. For higher incomes, it calculates the taxable income within each bracket and applies the corresponding tax rate. If the income is negative, it displays "Invalid income." The code prints the employee's name, calculated income tax, and error messages, if any. This program demonstrates how conditional logic can be used to implement a progressive tax calculation system, ensuring that the tax amount is calculated accurately based on income levels.

Q3:

Input	Processing	Output
• ID • Name • Salary • Performance Rating • Company's Performance	• Prompt for companyPerformance • GET companyPerformance • Read record from file • Calculate employeeBonus • Display Name, employeeBonus, msg	• Name • Employee Bonus • msg

[Pseudocode]:

```
1.  SET msg = ""
2.  SET personalSalaryMultipler = 0
3.  SET companySalaryMultipler = 0
4.  Prompt for companyPerformance
5.  GET companyPerformance
6.  READ first record
7.  DOWHILE NOT EOF (End of File)
8.  IF (performanceRating == 1)
        personalSalaryMultipler = 0.8

    ELSE IF (performanceRating == 2)
        personalSalaryMultipler = 1

    ELSE IF (performanceRating == 3)
      personalSalaryMultipler = 1.5

    ELSE IF (performanceRating == 4)
      personalSalaryMultipler = 2

      ELSE
      msg = "Invalid Performance Rating"
      ENDIF
    ENDIF
   ENDIF
  ENDIF
 ENDIF

9.  IF (companyPerformance < 100)
        companySalaryMultipler = 0.8
```

```
ELSE IF (companyPerformance >= 100 && companyPerformance <= 105)
   companySalaryMultipler = 1

 ELSE IF (companyPerformance > 105)
   companySalaryMultipler = 1.5

  ELSE
   msg = "Invalid Company Performance Rating"
  ENDIF
 ENDIF
ENDIF
ENDIF
```

10. Calculate employeeBonus
11. DISPLAY employeeID, employeeName, employeeBonus, msg
12. READ next record

    ENDDO

[Desk Checking]:

1. Input Data

	First Record	Second Record	Third Record
ID	A0001	B0002	C0003
Name	Jake	Scully	Terry
Salary	70000	50000	95000
Performance Rating	3	0	4
Company's Performance	100	95	110

2. Expected Results

	First Data Set	Second Data Set	Third Data Set
Name	Jake	Scully	Terry
Employee Bonus	105000	0	285000
Msg		Invalid Performance Rating	

3. Desk Check Table

Statement No.	ID	Name	Salary	Performance Rating	Company Performance	Employee Bonus	DOWHILE	Msg
First Pass								
1,2,3								
4,5					100			

6,7	A0001	Jake	70000	3			NOT EOF	
8,9								
10						105000		
6,7	B0002	Scully	50000	0			NOT EOF	
8,9								
10						0		Invalid Perfor mance Rating
6,7	C0003	Terry	95000	4			NOT EOF	
8,9								
10						190000		
11	A0001 B0002 C0003	Jake Scully Terry	70000 50000 95000	3 0 4		105000 0 190000		Invalid Perfor mance Rating

Test Case 1,2 & 3:

```
Output - SalaryMultiplierCalculator (run)
run:
Please provide the company performance in %: 100
A0001   Jake    105000
B0002   Scully  0       Invalid Performance Rating
C0003   Terry   190000
BUILD SUCCESSFUL (total time: 1 second)
Terminal - localhost   Output
```

```
Output - SalaryMultiplierCalculator (run)
run:
Please provide the company performance in %: 95
A0001   Jake    84000
B0002   Scully  0       Invalid Performance Rating
C0003   Terry   152000
BUILD SUCCESSFUL (total time: 1 second)
Terminal - localhost   Output
```

Output - SalaryMultiplierCalculator (run)

```
run:
Please provide the company performance in %: 110
A0001   Jake    157500
B0002   Scully  0       Invalid Performance Rating
C0003   Terry   285000
BUILD SUCCESSFUL (total time: 1 second)
```

Terminal - localhost Output

SalaryMultiplierData - Notepad

File Edit Format View Help

```
A0001 Jake 70000 3
B0002 Scully 50000 0
C0003 Terry 95000 4
```

Source History

```java
import java.util.Scanner;
import java.io.*;
public class SalaryMultiplierCalculator {

    public static void main(String[] args) throws Exception {
        Scanner input = new Scanner(new File(pathname: "D:\\Study Materials\\Kaplan\\Term 3\\Introduction to Computer Science\\SalaryMultiplierCalculator\\SalaryMultiplierData.txt"));
        Scanner scanner = new Scanner(source: System.in);
        String employeeID, employeeName, msg;
        double personalSalaryMultipler;
        double companySalaryMultipler;
        int originalSalary;
        int employeeBonus;
        int performanceRating;
        int companyPerformance;

        System.out.print(s: "Please provide the company performance in %: ");
        companyPerformance = scanner.nextInt();

        while(input.hasNext()){
            msg = "";
            personalSalaryMultipler = 0;
            companySalaryMultipler = 0;
            employeeID = input.next();
            employeeName = input.next();
            originalSalary = input.nextInt();
            performanceRating = input.nextInt();


        if (performanceRating == 1){
            personalSalaryMultipler = 0.8;
        }
        else if (performanceRating == 2){
            personalSalaryMultipler = 1;
        }
        else if (performanceRating == 3){
            personalSalaryMultipler = 1.5;
        }
        else if (performanceRating == 4){
            personalSalaryMultipler = 2;
        }
        else {
            msg = "Invalid Performance Rating";
        }

        if (companyPerformance < 100){
            companySalaryMultipler = 0.8;
        }
        else if (companyPerformance >= 100 && companyPerformance <= 105 ){
            companySalaryMultipler = 1;
```

Terminal - localhost

```
28
29          if (performanceRating == 1){
30              personalSalaryMultipler = 0.8;
31          }
32          else if (performanceRating == 2){
33              personalSalaryMultipler = 1;
34          }
35          else if (performanceRating == 3){
36              personalSalaryMultipler = 1.5;
37          }
38          else if (performanceRating == 4){
39              personalSalaryMultipler = 2;
40          }
41          else {
42              msg = "Invalid Performance Rating";
43          }
44
45          if (companyPerformance < 100){
46              companySalaryMultipler = 0.8;
47          }
48          else if (companyPerformance >= 100 && companyPerformance <= 105 ){
49              companySalaryMultipler = 1;
50          }
51          else if (companyPerformance > 105){
52              companySalaryMultipler = 1.5;
53          }
54          else{
55              msg = "Invalid Company Performance Rating";
56          }
57
58          employeeBonus = (int) (originalSalary * personalSalaryMultipler * companySalaryMultipler);
59          System.out.printf(format: "%s \t%s \t%d \t%s \n", args: employeeID, args: employeeName, args: employeeBonus, args: msg);
60          }
61
62      }
63  }
64
```

Terminal - localhost

Algorithm:

```
import java.util.Scanner;
import java.io.*;
public class SalaryMultiplierCalculator {

    public static void main(String[] args) throws Exception {
        Scanner input = new Scanner(new File("D:\\Study Materials\\Kaplan\\Term 3\\Introduction to
Computer Science\\SalaryMultiplierCalculator\\SalaryMultiplierData.txt"));
        String employeeID, employeeName, msg;
```

```
double personalSalaryMultipler;
double companySalaryMultipler;
int originalSalary;
int employeeBonus;
int performanceRating;
int companyPerformance;

while(input.hasNext()){
   msg = "";
   personalSalaryMultipler = 0;
   companySalaryMultipler = 0;
   employeeID = input.next();
   employeeName = input.next();
   originalSalary = input.nextInt();
   performanceRating = input.nextInt();
   companyPerformance = input.nextInt();

if (performanceRating == 1){
   personalSalaryMultipler = 0.8;
}
else if (performanceRating == 2){
   personalSalaryMultipler = 1;
}
else if (performanceRating == 3){
   personalSalaryMultipler = 1.5;
}
else if (performanceRating == 4){
   personalSalaryMultipler = 2;
}
else {
   msg = "Invalid Performance Rating";
}


if (companyPerformance < 100){
   companySalaryMultipler = 0.8;
}
else if (companyPerformance >= 100 && companyPerformance <= 105 ){
   companySalaryMultipler = 1;
}
else if (companyPerformance > 105){
   companySalaryMultipler = 1.5;
}
else{
```

```
            msg = "Invalid Company Performance Rating";
        }

        employeeBonus = (int) (originalSalary * personalSalaryMultipler * companySalaryMultipler);
        System.out.printf("%s \t%s \t%d \t%s \n", employeeID, employeeName, employeeBonus, msg);
        }
    }
}
```

Code Logic:

This code reads employee data from a file and calculates bonuses based on performance and company performance ratings. It uses a loop to process each employee's information. For performance rating, it assigns personal salary multipliers ranging from 0.8 to 2, with an "Invalid Performance Rating" message for other values. For company performance, it assigns company salary multipliers between 0.8 and 1.5. If the performance ratings or company performance are invalid, error messages are sent. Finally, it calculates the employee's bonus based on the original salary and both multipliers, then prints the results for each employee.

This code demonstrates how to use conditional statements to handle different scenarios and calculate bonuses based on specified rules. It also highlights the importance of input validation and feedback for error handling.